Worksheet: Second-Order Conditions

ECON 441: Introduction to Mathematical Economics — Instructor: Div Bhagia

The second derivative of a function $f(x)$, written $f''(x)$, is the derivative of $f'(x)$.

On Monday you found the critical points $x_1^* = 2$ and $x_2^* = 6$ of

$$f(x) = x^3 - 12x^2 + 36x + 8$$

Find $f''(x)$. What are the values of $f''(x_1^*)$ and $f''(x_2^*)$? What does the second-derivative test tell you about each critical point?

Is f concave, convex, or neither (a) on the whole real line, (b) for $x \geq 4$?

Given the quadratic profit function

$$\pi(Q) = hQ^2 + jQ + k$$

what parameter restrictions are called for to reflect the following assumptions?

(a) If nothing is produced, the profit will be negative (because of fixed costs).

(b) The profit function is strictly concave ($\pi''(Q) < 0$ for all Q).

(c) The maximum profit occurs at a positive output level Q^*.